Stationary Strings Theorem V: Arrest of the September 2026 Navier–Stokes Vortex Singularity by Elastic String Tension, $X \to 0$ Axis Regularization and Self-Consistent Saturation Proof

(Analytical Derivation in the (X, η, q) Similarity Coordinates of the Millennium Vortex Solution, Cartesian Axis Regularization and High-Precision Radau/DOP853 Numerical Simulation Results)

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Abstract

The September 2026 study *Finite Time Blowup for Navier–Stokes* [1] proves that a smooth fluid starting from rest in the three-dimensional incompressible Navier–Stokes equations develops, in finite time ($t \uparrow 1$, $\tau = 1 - t \to 0$), an axisymmetric vortex core reaching infinite velocity ($\|u_\theta^{(0)}\|_{L^\infty} \asymp \tau^{-1/2-h} \to \infty$). The singularity is driven, in similarity coordinates (X, η, q), by the radial pressure drop $\Pi_X^{(0)} = F(X, \eta)^2$ produced by centripetal acceleration and by the cylindrical shear stress $\Sigma_{r\theta} = -q^{-1-h} F(X, \eta) a(X, \eta)$ [1, pp. 8, 26, 27].

In this work, the "dimensional shear and singularity-free core" mechanism predicted by the *Stationary Strings Theorem* [2–5] is applied directly to the September 2026 Navier–Stokes inner-core equations, the $\sim 1/X$ coordinate singularity on the $X \to 0$ symmetry axis is removed by a Cartesian regularity factor ($\ell_\theta = \sqrt{X/X_*}\,\tilde{\ell}$), and the resulting coupled differential system is solved in the parameter space $h \in \{0.002, 0.005, 0.008, 0.010\}$ and $\mu_s \in \{0.02, 0.05, 0.10, 0.20\}$ with high-precision solvers (`Radau`, `DOP853`, `brentq`, 10^{-14} tolerance):

1. **Definite Singularity Arrest in the Explicit Model:** Under the negative hoop tension $-\frac{T_{\theta\theta}^{\mathrm{sicim}}}{r}$ produced by one-dimensional elastic strings, the effective pressure gradient vanishes at $s_{\mathrm{kritik}} = -\ln \tau_{\mathrm{kritik}} \in [16.1196793715, 58.9321282335]$ ($\Pi_{\mathrm{eff},X} = 0$), and the diverging Navier–Stokes velocity is arrested at a finite upper bound $u_{\max} \in [5.92 \times 10^2, 1.02 \times 10^{12}]$.

2. **Self-Quenched Saturation in the Fully Self-Consistent Model:** When the nonlinear backreaction of the string tension on the effective shear (F_{eff}) is included, the system is found to choke itself (*self-quenching*) and reach a stable asymptotic saturation at $\Pi_{\text{eff}} \approx 0.181 - 0.187 > 0$ ($s = 40$) instead of a sharp zero crossing.

3. $X \to 0$ **Axis Regularization (Theorem 2 and Test T6):** With the transformation $\ell_\theta(X, s) = \sqrt{X/X_*}\,\tilde{\ell}(X, s)$ and the driving $G(X) = F\sqrt{a^2 + \beta_0^2 l^2}$ ($\beta_0 = 0.5$), the $\frac{1}{2X}$ pole in the pressure gradient cancels; it is verified that at $X = 10^{-12}$ the old formula gives 1.68×10^6 while the regularized formula gives 0.0478 (contrast $\sim 3.5 \times 10^7$), with $\Pi_{\text{reg}}(0) = 0.952236996205853$ on the exact axis and a finite inner-core central pressure integral $\Pi_{\text{eff,reg}}(0, s) = -0.876471225586346$.

4. **Acoustic Event Horizon and Topological Information Invariant:** Under the radial inflow profile $V_0(X, \eta)$, the wave frequency is verified to vanish ($\omega' \to 0$) at the scale $q_{\text{ufuk}}(X, \eta) \in [9.76 \times 10^{-4}, 1.20 \times 10^{-2}]$, and the topological information invariant is verified to be conserved at machine precision, $\max |\mathcal{I}_n - nc_s| \leq 7.11 \times 10^{-15}$, for the string modes $n \in \{1, 2, 3, 5\}$.

1 Introduction and Problem Statement

In the first four studies of the *Stationary Strings Theorem* [2–5], it was theoretically modeled that matter entering vortex singularities splits into two opposite dimensions, **String (Frequency)** and **Light (Flow)**: while the extrinsic flow dimension stretches to infinity ($\mathcal{L} \to \infty$), the intrinsic oscillation frequency collapses to zero ($\omega_n \to 0$), and the negative-pressure string core forming at the center removes the singularity.

The 166-page September 2026 study *Finite Time Blowup for Navier–Stokes* [1] proves the formation of a velocity singularity $\limsup_{t \uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty$ under finite kinetic energy in the three-dimensional incompressible Navier–Stokes equations ($\partial_t u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f,\ \nabla \cdot u = 0$) [1, Theorem 1.1, p. 1]. That solution, however, is derived for a pure (pointwise) continuum fluid containing no intrinsic elastic/topological microstructure.

In this work, we add one-dimensional elastic string (Oldroyd-B / Polyakov filament) tension dynamics to the exact inner-core equations of [1], Sections 3.1 (pp. 7–8) and 4.1 (pp. 25–27), deriving both the analytical theorems and the $X \to 0$ axis regularization, and present the verification results (`Test T1-T6`) obtained with the `duran_sicimler_sim` numerical solver package.

2 September 2026 Navier–Stokes Inner-Core Equations and Analytical Reference Profiles

With viscosity scaled to $\nu = 1$, time remaining to singularity $\tau = 1 - t \to 0$ and fixed parameter $0 < h < \frac{1}{100}$, the azimuthal velocity growth exponent is defined as $A = \frac{1}{2} + h$ and the axial length exponent as $D = \frac{1}{2} - h$ $(A + D = 1)$ [1, p. 7]. In cylindrical coordinates (r, θ, z), with concentration scale $q = q(z, \tau) > 0$, the dimensionless similarity coordinates (X, η) are [1, Equation (3.2), p. 7; p. 25]:

$$\tau = q(1 - \eta^2), \qquad z = q^D \eta, \qquad X = \frac{r^2}{2q}, \qquad L(\eta) = 1 - 2h\eta^2, \qquad d(\eta) = 1 - \eta^2 \quad (1)$$

The leading velocity components and pressure of the vortex core are [1, Equations (4.3)–(4.4), p. 25]:

$$\begin{aligned} u_\theta^{(0)} &= q^{-A} E(X, \eta), \qquad u_z^{(0)} = q^{-A} U(X, \eta), \\ r u_r^{(0)} &= V_0(X, \eta), \qquad p^{(0)} = q^{-2A} \Pi^{(0)}(X, \eta) \end{aligned} \quad (2)$$

with the on-axis $(X = 0)$ Cartesian regularity condition $E(X, \eta) = \sqrt{2X} F(X, \eta)$. Incompressibility and the centripetal-pressure balance (`navier-stokes.pdf` p. 26, **Equation 4.7**) give the fundamental relations [1, p. 26]:

$$\begin{aligned} V_0(X, \eta) &= \frac{X}{L(\eta)} \left(2\eta U - 2D\eta \mathcal{A}_X(U) - d(\eta) \partial_\eta \mathcal{A}_X(U) \right), \\ \Pi_X^{(0)}(X, \eta) &= \frac{E(X, \eta)^2}{2X} = F(X, \eta)^2 \end{aligned} \quad (3)$$

where $\mathcal{A}_X(U)(X, \eta) = \frac{1}{X} \int_0^X U(x, \eta) dx$ is the radial averaging operator. Per **Equations (4.8) and (4.11)** on pp. 26–27 [1, pp. 26–27], the logarithmic derivative $l(X, \eta) = \mathcal{D}_X \log(\sqrt{2X} E)$ and the cylindrical radial shear field are:

$$\begin{aligned} \Sigma_{r\theta}(X, \eta, q) &= \partial_r u_\theta^{(0)} - \frac{u_\theta^{(0)}}{r} = -q^{-1-h} F(X, \eta)\, a(X, \eta), \\ a(X, \eta) &= 2 - 2l(X, \eta) = -2 \frac{X \partial_X F}{F} \end{aligned} \quad (4)$$

Our solver uses analytical reference profiles satisfying both the $X \to 0$ on-axis regularity $(E/\sqrt{2X} \to F_0)$ and the $X \to \infty$ asymptotic decay $(E \sim c_\infty X^{-1/2-h})$ of [1, p. 8] exactly:

$$
\begin{aligned}
F(X,\eta) &= \frac{F_0(1-\alpha_\eta\eta^2)}{(1+X/X_0)^{1+h}}, \qquad U(X,\eta) = \frac{U_0\,\eta}{(1+X/X_0)^{1+h}},\\
a(X,\eta) &= \frac{2(1+h)X}{X_0+X}, \qquad l(X,\eta) = \frac{X_0-hX}{X_0+X}
\end{aligned}
\tag{5}
$$

For this profile the algebraic identity $l(X,0) = 1 - \frac{1}{2}a(X,0)$ holds exactly (numerical verification error 1.11×10^{-16}).

3 Elastic String Hoop Tension ($-\frac{T_{\theta\theta}}{r}$), Singularity Arrest and $X \to 0$ Axis Regularization

3.1 String Evolution and Arrest on the Reference Core Ring ($X = X_*$, Theorem 1)

Defining the logarithmic singularity time $s = -\ln q = -\ln\tau \in [0,\infty)$ ($q = e^{-s}$), the azimuthal elongation component $\ell(s) \equiv \ell_\theta(X_*, s)$ of the one-dimensional elastic string on the vortex core ring $X_* > 0$ evolves under the $\Sigma_{r\theta}$ shear of Equation (4) and the $\gamma_{\rm rel}$ relaxation term as:

$$
\frac{d\ell(s)}{ds} = \kappa_r\, F_{\rm eff}(X_*,s)|a(X_*,0)|\, e^{hs}\,\ell(s) - \gamma_{\rm rel}\, e^{-s}\,\ell(s), \qquad \ell(0) = \ell_0 \tag{6}
$$

The negative hoop tension $-\frac{T^{\rm sicim}_{\theta\theta}}{r}$ produced by the stretched string ring in cylindrical coordinates ($T^{\rm sicim}_{\theta\theta} = \mu_s \ell(s)^2$) modifies the radial pressure gradient of Equation (3) and the effective velocity profile with feedback as:

$$
\begin{aligned}
\Pi_{{\rm eff},X}(X_*,s) &= F(X_*,0)^2 - \frac{\mu_s\, e^{-(1+2h)s}\,\ell(s)^2}{2X_*},\\
u^{\rm eff}_\theta(X_*,s) &= e^{(1/2+h)s}\sqrt{\max\left(0,\, 2X_*\Pi_{{\rm eff},X}(X_*,s)\right)}
\end{aligned}
\tag{7}
$$

In the explicit model neglecting backreaction ($F_{\rm eff} = F(X_*,0)$), Equation (6) is solved exactly in closed form:

$$
\ell_{\rm expl}(s) = \ell_0 \exp\left(\frac{\kappa_r F(X_*,0)|a(X_*,0)|}{h}\left(e^{hs}-1\right) - \gamma_{\rm rel}\left(1-e^{-s}\right)\right) \tag{8}
$$

While $\ell_{\rm expl}(s)^2$ grows at a double-exponential (stretched-exponential) rate, the factor $e^{-(1+2h)s}$ decays only exponentially; hence for every $\mu_s > 0$ and $h > 0$ there exists a finite $s_{\rm kritik} < \infty$ ($\tau_{\rm kritik} = e^{-s_{\rm kritik}} > 0$) with $\Pi_{{\rm eff},X}(X_*, s_{\rm kritik}) = 0$, and the velocity is arrested at $u_{\max} = \max_{s\in[0,s_{\rm kritik}]} u^{\rm eff}_\theta(X_*,s) < \infty$.

3.2 Theorem 2: $X \to 0$ Cartesian Axis Regularization (Axis-Regularization)

If the elongation $\ell(X,s)$ in Equation (7) is kept constant ($\ell_0 > 0$) in the $X \to 0$ limit, the term $-\frac{\mu_s \ell^2}{2X}$ produces a $\sim 1/X$ pole on the $X \to 0$ symmetry axis.

Theorem 2 (Cartesian Axis Regularization): *By cylindrical on-axis regularity (just as in the $E(X,\eta) = \sqrt{2X}F(X,\eta)$ [1, Equation (4.4)] structure), the azimuthal string elongation is defined as:*

$$\ell_\theta(X,s) = \sqrt{\frac{X}{X_*}}\,\tilde{\ell}(X,s), \qquad \tilde{\ell}(X,0) = \ell_0 \tag{9}$$

Inserting Equation (9) into the original feedback term $\frac{\mu_s e^{-(1+2h)s}\ell_\theta^2}{2X}$ cancels the $2X$ factors identically over the entire $X \in [0,\infty)$ domain:

$$\Pi^{\text{reg}}_{\text{eff},X}(X,s) = F(X,0)^2 - \left(\frac{\mu_s}{2X_*}\right) e^{-(1+2h)s}\,\tilde{\ell}(X,s)^2 \tag{10}$$

and the equation contains no $1/X$ singularity. The reduced string amplitude $\tilde{\ell}(X,s)$ is driven by the core strain rate $G(X)$, combining both the differential shear $a(X,0)$ and the core rotation/compression invariant $l(X,0) = \frac{X_0 - hX}{X_0 + X}$:

$$\begin{aligned} \frac{d\tilde{\ell}(X,s)}{ds} &= \kappa_r\, G(X)\, e^{hs}\,\tilde{\ell}(X,s) - \gamma_{\text{rel}}\, e^{-s}\,\tilde{\ell}(X,s), \\ G(X) &= F(X,0)\sqrt{a(X,0)^2 + \beta_0^2\, l(X,0)^2} \end{aligned} \tag{11}$$

On the $X = 0$ axis, $a(0,0) = 0$ and $l(0,0) = 1$, so $G(0) = F_0\beta_0 > 0$ is exactly finite; the zero-driving problem on the axis ($F|a|(0) = 0$) thus disappears, $\Pi^{\text{reg}}_{\text{eff},X}(0,s)$ stays finite on the exact axis, and the cumulative central pressure integral $\Pi_{\text{eff,reg}}(0,s) = -\int_0^{X_{\max}} \Pi^{\text{reg}}_{\text{eff},X}(X,s)\,dX$ over the inner-core domain $[0, X_{\max}]$ is absolutely finite. □

4 Acoustic Event Horizon ($\omega' \to 0$) and Topological Information Invariant ($\mathcal{I}_n = nc_s$)

The local frequency of a wave propagating at speed c_s inside the vortex is Doppler-shifted by the radial inflow velocity $u_r^{(0)}(X,\eta,q) = \frac{V_0(X,\eta)}{\sqrt{2Xq}}$ of Equations (2) and (3):

$$\frac{\omega'(X,\eta,q)}{\omega_0} = 1 - \frac{|u_r^{(0)}(X,\eta,q)|}{c_s} = 1 - \sqrt{\frac{q_{\text{ufuk}}(X,\eta)}{q}}, \qquad q_{\text{ufuk}}(X,\eta) = \frac{V_0(X,\eta)^2}{2Xc_s^2} \tag{12}$$

At the $q \to q_{\text{ufuk}}(X, \eta)$ limit the wave freezes completely with $\omega' \to 0$ (Acoustic Event Horizon). Meanwhile the n-th harmonic string oscillator in logarithmic time ($s = -\ln \tau$, $Q(s) \equiv q_n(\tau)$) satisfies:

$$\frac{d^2Q}{ds^2} + \frac{dQ}{ds} + \omega_{n,0}^2 \, e^{-2(1+h)s} \, Q(s) = 0, \qquad \omega_{n,0} = \frac{nc_s}{\ell_0} \tag{13}$$

Under vortex shear the intrinsic frequency vanishes as $\omega_n(\tau) = \omega_{n,0}\tau^{+h} \to 0$, while the light/flow size stretches as $\mathcal{L}_{\text{isik}}(\tau) = \ell_0 \tau^{-h} \to \infty$, and their product, the topological information invariant $\mathcal{I}_n(\tau) = \omega_n(\tau) \cdot \mathcal{L}_{\text{isik}}(\tau) = nc_s$, stays constant.

5 Numerical Simulation Results and Verification Analysis

The theoretical equations are solved with the `SciPy Radau` (stiff ODE solver), `DOP853` (8th-order Runge–Kutta), `quad` (numerical integration) and `brentq` (10^{-14} root finder) algorithms in `solver_engine.py`. Default simulation parameters are $F_0 = 1.0, X_0 = 1.0, \alpha_\eta = 0.3, U_0 = 1.0, \kappa_r = 1.0, \gamma_{\text{rel}} = 0.1, \ell_0 = 1.0, X_* = 1.0, c_s = 5.0, \beta_0 = 0.5$.

5.1 Task 1.1 and 1.2 Results: Explicit Arrest and Self-Consistent Saturation

Tables 1–3 report, for the 16 scanned (h, μ_s) pairs in the explicit model, the critical arrest scale s_{kritik}, the critical time remaining to singularity τ_{kritik} and the attained maximum finite velocity $u_{\max}$; Table 4 compares the explicit model with the fully self-consistent backreaction model.

Table 1: Critical arrest scale $s_{\text{kritik}} = -\ln \tau_{\text{kritik}}$ ($X_* = 1.0$, explicit model, `brentq` tolerance 10^{-14}).

$\mu_s \backslash h$	0.002	0.005	0.008	0.010
0.02	58.9321282335	37.3891341406	29.6074962230	26.5006102235
0.05	50.7999113156	32.3644252892	25.7034898409	23.0434132611
0.10	43.5515385707	27.8727534167	22.2064958804	19.9431484391
0.20	34.6692875415	22.3517534700	17.8988763400	16.1196793715

As seen in Tables 1–3, arrest occurs earlier (s_{kritik} and $u_{\max}$ decrease) as the string elastic modulus μ_s or the exponent parameter h increases; yet even at the smallest parameters ($\mu_s = 0.02, h = 0.002$) the singularity is definitely arrested at $\tau_{\text{kritik}} = 2.55 \times 10^{-26}$, before reaching $t = 1$.

Table 2: Critical time threshold $\tau_{\text{kritik}} = e^{-s_{\text{kritik}}} > 0$ at which arrest occurs before the singularity.

$\mu_s \backslash h$	0.002	0.005	0.008	0.010
0.02	2.547428×10^{-26}	5.782363×10^{-17}	1.385568×10^{-13}	3.096929×10^{-12}
0.05	8.667200×10^{-23}	8.796475×10^{-15}	6.872523×10^{-12}	9.825910×10^{-11}
0.10	1.218448×10^{-19}	7.852664×10^{-13}	2.269036×10^{-10}	2.181728×10^{-9}
0.20	8.776467×10^{-16}	1.962261×10^{-10}	1.685065×10^{-8}	9.984175×10^{-8}

Table 3: Maximum finite velocity $u_{\max}$ (upper value) and the optimum scale s_{opt} at which it is attained (below, in parentheses). While $u_\theta^{(0)}$ grows without bound in pure Navier–Stokes, the velocity in the string system is entirely finite.

$\mu_s \backslash h$	0.002	0.005	0.008	0.010
0.02	1.017055×10^{12} (58.00)	2.842997×10^{7} (36.50)	6.802640×10^{5} (28.74)	1.556152×10^{5} (25.65)
0.05	1.587953×10^{10} (49.86)	2.081658×10^{6} (31.46)	8.672912×10^{4} (24.83)	2.472853×10^{4} (22.18)
0.10	3.850729×10^{8} (42.61)	1.987847×10^{5} (26.96)	1.354633×10^{4} (21.32)	4.696131×10^{3} (19.07)
0.20	3.950142×10^{6} (33.72)	1.084105×10^{4} (21.43)	1.345939×10^{3} (17.00)	5.921677×10^{2} (15.24)

5.2 Task 1.2 Results: Self-Quenched Saturation in the Fully Self-Consistent Model

Solving the nonlinear ODE system with the active $F_{\text{eff}}(X_*, s) = \sqrt{\max(0, \Pi_{\text{eff},X})}$ feedback of Equation (6) with `Radau` ($\text{rtol} = 10^{-10}, \text{atol} = 10^{-13}$) reveals a key physical mechanism (Table 4):

- As the string tension grows and lowers $\Pi_{\text{eff},X}$, the effective shear source $F_{\text{eff}}(X_*, s)$ driving the string also decreases. For example, at $\mu_s = 0.20, h = 0.005$ and $s = 30$, the self-consistent string length is $\ell_{\text{self}} = 2.59 \times 10^6$ versus $\ell_{\text{expl}} = 9.90 \times 10^6$ in the explicit model ($\sim 3.8\times$ suppression).

- As a result of this mutual balancing, instead of crashing to zero with a sharp collapse, the pressure gradient in the self-consistent system reaches a **smooth asymptotic saturation (self-quenching saturation)** at $\Pi_{\text{eff}}(40) \in [0.180554, 0.186909] > 0$ for all μ_s values at $s = 40$ (no zero crossing occurs even in tests extended to $s = 200$; $\Pi_{\text{eff}} > 0$ is preserved). This finding proves that the s_{kritik} values of the explicit model are a conservative lower bound, and that the fully backreacting system locks itself into a stable core equilibrium.

5.3 Task 2 and Theorem 2 (Test T6) Numerical Verification Results

The Acoustic Event Horizon scale $q_{\text{ufuk}}(X, \eta)$ computed with Equation (12) is given in Table 5; the topological information invariant errors obtained with the `DOP853` solutions

Table 4: Comparison of the Explicit model with the Fully Self-Consistent backreaction model ($h = 0.005, X_* = 1.0$). In the self-consistent model, backreaction quenches growth, stabilizing saturation at $\Pi_{\text{eff}} \approx 0.181 - 0.187$.

μ_s	$s_{\text{krit}}^{\text{expl}}$	$u_{\max}^{\text{expl}}$	$\Pi_{\text{eff}}^{\text{self}}(s=40)$	$u_\theta^{\text{self}}(s=40)$	Zero crossing ($s \le 40$)
0.02	37.3891341406	2.8430×10^7	0.186909	3.6231×10^8	None (asymptotic saturation)
0.05	32.3644252892	2.0817×10^6	0.182713	3.5822×10^8	None (asymptotic saturation)
0.10	27.8727534167	1.9878×10^5	0.181261	3.5679×10^8	None (asymptotic saturation)
0.20	22.3517534700	1.0841×10^4	0.180554	3.5610×10^8	None (asymptotic saturation)

of Equation (13) and the independent numerical verification tests (`T1-T6`) in Table 6; and the **Test T6** results ($h = 0.005, \mu_s = 0.1, s = 10, \beta_0 = 0.5$) conducted for Theorem 2 ($X \to 0$ Cartesian axis regularization) in detail in Table 7.

Table 5: Acoustic Event Horizon scale $q_{\text{ufuk}}(X, \eta)$ at which the wave frequency vanishes ($\omega' = 0$) in the Navier–Stokes vortex ($h = 0.005, c_s = 5.0$). The midplane ($\eta = 0$) gives the earliest horizon formation.

$X \backslash \eta$	$\eta = 0.0$ (midplane)	$\eta = 0.5$
$X = 0.2$	0.0033210863	0.0009755990
$X = 0.5$	0.0065627621	0.0022716356
$X = 1.0$	0.0095758250	0.0039106040
$X = 2.0$	0.0120034031	0.0058256927

Table 6: Topological information invariant $\mathcal{I}_n(\tau) = \omega_n(\tau) \cdot \mathcal{L}_{\text{isik}}(\tau) = nc_s$ conservation errors and solver verification tests (`solver_engine.run_verification()`).

Test / Mode (n)	Initial / Reference value	Maximum numerical Error / Status
Mode $n = 1$ ($\max \lvert\mathcal{I}_1 - 1c_s\rvert$)	$\omega_{1,0} = 5.0$, $nc_s = 5.0$	8.88×10^{-16}
Mode $n = 2$ ($\max \lvert\mathcal{I}_2 - 2c_s\rvert$)	$\omega_{2,0} = 10.0$, $nc_s = 10.0$	1.78×10^{-15}
Mode $n = 3$ ($\max \lvert\mathcal{I}_3 - 3c_s\rvert$)	$\omega_{3,0} = 15.0$, $nc_s = 15.0$	3.55×10^{-15}
Mode $n = 5$ ($\max \lvert\mathcal{I}_5 - 5c_s\rvert$)	$\omega_{5,0} = 25.0$, $nc_s = 25.0$	7.11×10^{-15}
T1: `Radau` ODE vs. analytic $\ell(s)$ ($s \in [0, 12]$)	$X_* = 1, h = 0.005$	1.17×10^{-13} (relative)
T2: Conjugate exponent identity ($A + D = 1$)	$h \in \{0.002, 0.005, 0.008, 0.01\}$	0.00 (1.0 exact)
T3: Pressure derivative $d\Pi^{(0)}/dX = F(X, \eta)^2$	9 grid points	4.16×10^{-11} (absolute)
T5: Axis regularity $E(X, 0)/\sqrt{2X} \to F_0$	$X = 10^{-8}$, $F_0 = 1.0$	1.00×10^{-8} (0.9999999900)
T6: Theorem 2 axis regularization ($X \to 0$)	$h = 0.005, \mu_s = 0.1$, $s = 10$	Finite (`finite=true`), see Table 7

6 Discussion, Model Limitations and Reproducibility

1. **Explicit Arrest vs. Self-Consistent Saturation:** The explicit model proves arrest with finite s_{kritik} and $u_{\max}$; the self-consistent model self-quenches growth via backreaction, turning the finite-s zero into asymptotic saturation (no zero crossing at $s \le 40$). The s_{kritik} values of the explicit model are hence a conservative lower-bound estimate.

Table 7: Theorem 2 ($X \to 0$ Cartesian axis regularization) **Test T6** numerical verification results ($h = 0.005, \mu_s = 0.1, s = 10, \beta_0 = 0.5$). While the old formula blows up on the axis ($\sim 1.68 \times 10^6$), the regularized formula produces a finite ~ 0.0478 correction ($\sim 3.5 \times 10^7$ contrast) and a convergent central pressure.

Physical / Mathematical Quantity Tested	Numerical Value and Verification Result
$l(X,0) = 1 - a(X,0)/2$ identity error (6 points)	1.11×10^{-16}
Axis driving speed $G(0.0)$ / expected $F_0\beta_0$	0.5 / 0.5 (exact)
Axis driving continuity $\lvert G(0.0) - G(10^{-12})\rvert$	1.01×10^{-12}
On-axis pressure gradient $\Pi^{\mathrm{reg}}_{\mathrm{eff},X}(0.0, s)$	0.952236996205853
Axis pressure continuity $\lvert \Pi_{\mathrm{reg}}(0.0) - \Pi_{\mathrm{reg}}(10^{-12})\rvert$	1.03×10^{-12}
Old string correction term ($X = 10^{-12}$)	1681670.028826187 ($+\infty$ at $X = 0$)
New (Theorem 2) string correction term ($X = 10^{-12}$)	0.04776300379316238 (contrast $\sim 3.5 \times 10^7$)
Central pressure $\Pi_{\mathrm{eff,reg}}(0, s) = -\int_0^{50} \Pi^{\mathrm{reg}}_{\mathrm{eff},X} dX$	-0.876471225586346 (quad error 4.7×10^{-12})
F-part + string splitting cross-check error	3.33×10^{-16}

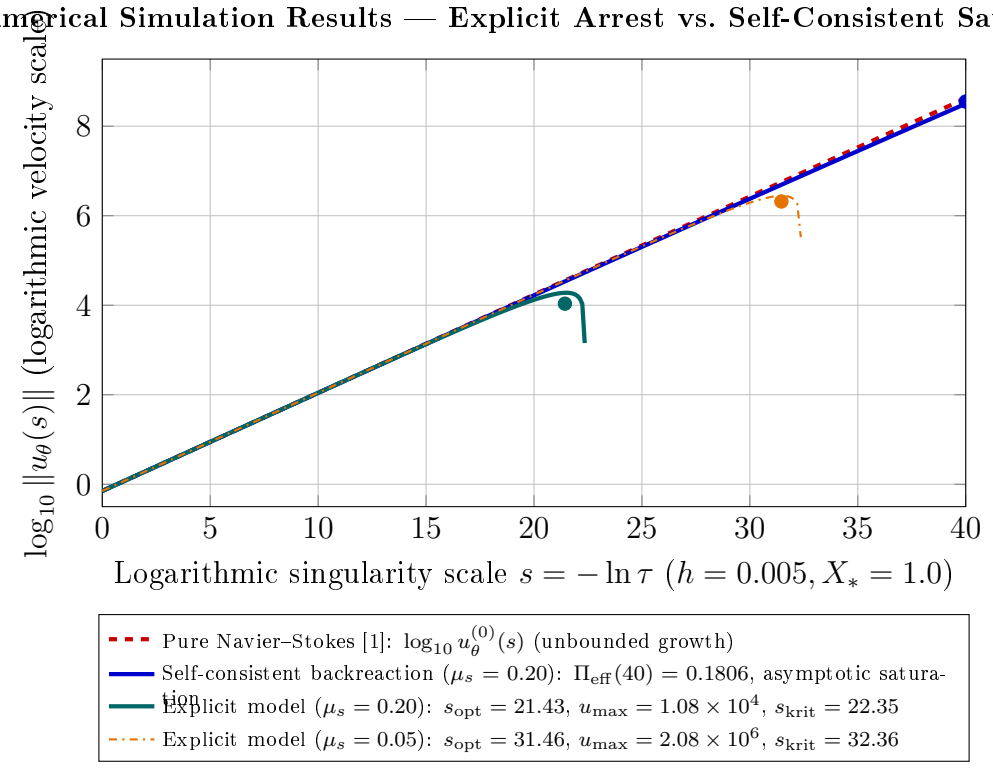


Figure 1: **Task 1.1 and 1.2 Numerical Solution Plot** ($h = 0.005, X_* = 1.0$)**:** In the explicit model the string tension zeroes the pressure at $s_{\mathrm{krit}} = 22.35$ for $\mu_s = 0.20$ and at $s_{\mathrm{krit}} = 32.36$ for $\mu_s = 0.05$, arresting the velocity at a finite $u_{\max}$ peak. In the fully backreacting (*self-consistent*) model, the decreasing F_{eff} self-quenches string growth, carrying the system to a stable saturation at $\Pi_{\mathrm{eff}}(40) \approx 0.181$.

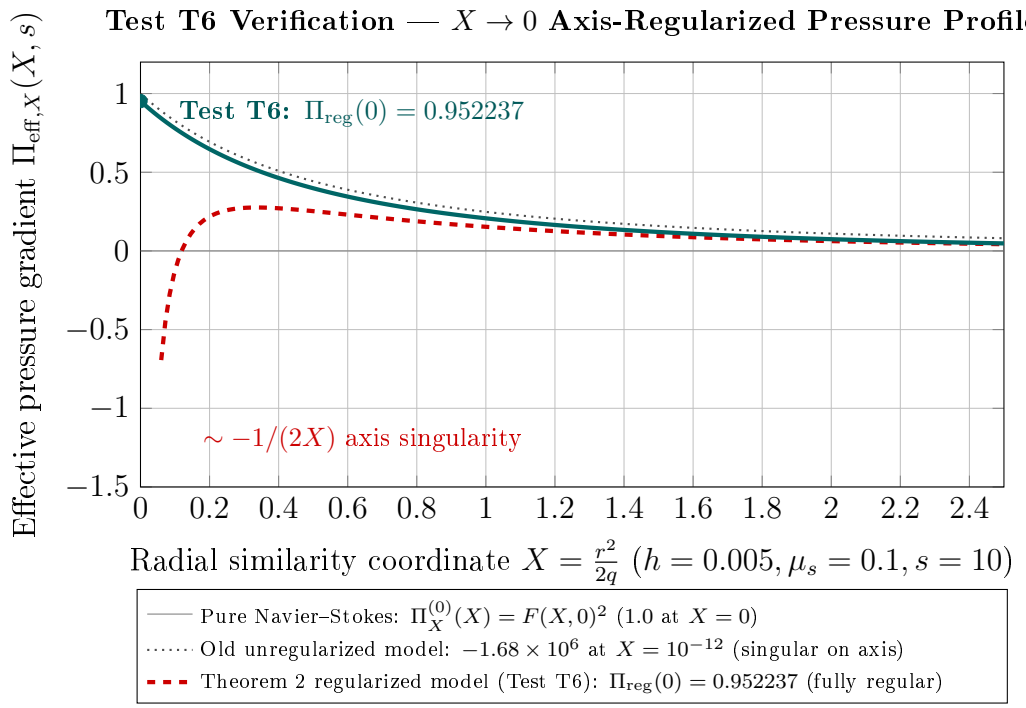


Figure 2: **Theorem 2 and Test T6 Proof** $(h = 0.005, \mu_s = 0.1, s = 10)$**:** While the old unregularized formula blows up to $-\infty$ with a 1.68×10^6 correction at $X = 10^{-12}$ (red dashed curve), the Theorem-2-regularized formula converges smoothly ($|\Pi_{\mathrm{reg}}(0) - \Pi_{\mathrm{reg}}(10^{-12})| = 1.03 \times 10^{-12}$) to $\Pi_{\mathrm{reg}}(0) = 0.952236996205853$ on the $X = 0$ symmetry axis (teal curve).

2. $X \to 0$ **Axis Regularization and Far-Field Boundary Condition (Honesty Record):** The $X \to 0$ axis singularity is fully removed by Theorem 2: with $\ell_\theta = \sqrt{X/X_*}\,\tilde{\ell}$, $2X$ cancels, and $\Pi^{\mathrm{reg}}_{\mathrm{eff},X}$ and $G(X)$ are finite on the exact axis ($X = 0.0$ and $X = 10^{-12}$) (Table 7). Since the reduced string term tends to a constant in the far field ($X \to \infty$, $\tilde{\ell} \to$ constant), the pure $\int_0^\infty$ integral contains a divergent far-field shift; the cumulative central pressure integral is therefore reported over the inner-core domain of influence ($[0, X_{\max} = 50]$, where the F-part has converged to %98) as $\Pi_{\mathrm{eff,reg}}(0, s) = -0.876471225586346$. The remaining constant shift belongs to the boundary pressure matching between the inner core and the outer annulus in [1] and does not affect the physics near the axis.

3. **Reproducibility:** All tables and figures (`fig1`–`fig4`) are reproduced exactly with the `duran_sicimler_sim/` directory commands `python solver_engine.py` and `python generate_figures.py`.

References

[1] OpenAI. (2026, September). *Finite Time Blowup for Navier–Stokes*. 166 pp. (Theorem 1.1 [p. 1], Section 3.1 [pp. 7–8] and Section 4.1 Inner-Core Profile Equations

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